

# Talking to a Robot’s Hidden State:

## Natural Language Autoencoders on a Vision–Language–Action Backbone

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### Abstract

Vision–language–action (VLA) policies such as NVIDIA’s GR00T expose a high-dimensional Qwen3-VL backbone hidden state  $\mathbf{h}$  that is causally upstream of the action head, yet there is no agreed-upon way to read  $\mathbf{h}$  out as natural language or to write back into it from language. We adapt the recently proposed Natural Language Autoencoder (NLA) recipe (Fraser-Taliente et al., 2026) to this setting and release **nla-groot**, an open implementation that (i) extracts per-token activations from GR00T’s backbone, (ii) trains a paired *activation verbalizer* AV ( $\mathbf{h} \rightarrow y$ ) and *activation reconstructor* AR ( $y \rightarrow \hat{\mathbf{h}}$ ) with joint supervised fine-tuning plus an optional GRPO phase that uses AR as a learned reconstruction reward, and (iii) injects AR’s output back into the live policy server as a backbone steering perturbation in LIBERO closed-loop simulation. We propose a three-axis evaluation harness—offline reconstruction, multimodal LLM grounding judge, and closed-loop steerability A/B—and report a *negative-result-rich* pilot on LIBERO Goal: our V3 checkpoint attains a healthy retrieval margin (0.124) and 100% caption *appropriateness* but only 8.3% anti-template specificity, 41.7% grounding, and a zero correct-minus-wrong steering gap on the single tested task. Reconstruction-only metrics would have called this a success; the harness does not. We argue this evaluation pattern, and the “shorthand template collapse” failure mode it surfaces, are a prerequisite for any future NLA-style interpretability work on VLA policies.

## 1 Introduction

Recent vision–language–action (VLA) models pair a Vision–Language backbone with a diffusion or flow-matching action head and act end-to-end on physical robots (NVIDIA, 2025; Brohan et al., 2023; Kim et al., 2024). As capabilities scale, an old question returns in a new form: *what does the model think it is about to do, expressed in a representation a human can audit and edit?* Mechanistic interpretability tooling for language models—probes (Alain and Bengio, 2017; Belinkov, 2022), sparse autoencoders (Cunningham et al., 2023; Bricken et al., 2023), and activation steering (Turner et al., 2023; Zou et al., 2023)—has not yet been adapted to the policy setting, where downstream behavior is itself an evaluation modality.

**The NLA recipe, on a VLA.** Fraser-Taliente et al. (2026) showed that an LM’s hidden state can be “round-tripped” through natural language: a small *activation verbalizer* (AV) is trained to caption an injected activation, and a paired *activation reconstructor* (AR) is trained to invert the caption back into activation space, providing both an interpretability artifact (the caption) and a causal handle (the reconstructed vector). This paper asks whether the same recipe transfers to a VLA backbone, where the hidden state is intended to carry scene-and-task semantics that the action head consumes.

### Contributions.

1. **System:** an end-to-end open implementation (`nla-groot`) that extracts GR00T backbone activations (Qwen3-VL, 16-layer truncation,  $d=2048$ ), labels them with a multimodal teacher, jointly SFT-trains AV/AR on top of a Qwen3-4B-Instruct base, and exposes AR’s output as a live backbone steering vector in the official GR00T LIBERO policy server (§3).
2. **Evaluation harness:** three complementary axes—offline reconstruction (FVE/cosine/retrieval), multimodal LLM judge on cached frames (axes B/C), and closed-loop sim A/B with matched-, mismatched-, and null-caption arms—wired into a single PASS/WARN/FAIL *scorecard* (§4).
3. **Pilot findings:** on a LIBERO 4-suite combined corpus, our best checkpoint passes reconstruction-style metrics but *fails* grounding and steering. We characterize this “shorthand template collapse” failure mode, show that retrieval margin and reconstruction cosine alone are insufficient as a workshop-grade success criterion, and surface a train–deploy distribution gap that limits behavioral steering (§5).

**Thesis.** *NLA-style autoencoding is a credible direction for VLA interpretability, but only behind a multi-axis harness; reconstruction metrics in isolation are misleading.*

## 2 Background and related work

**Vision–language–action policies.** GR00T (NVIDIA, 2025) couples a Cosmos-Reason2-2B / Qwen3-VL backbone with a diffusion action head; only the first `SELECT_LAYER=16` decoder layers are retained at deployment, so the policy effectively reads a 2048-d hidden state at layer 16. Related VLAs include RT-2 (Brohan et al., 2023), OpenVLA (Kim et al., 2024), and Octo (Octo Model Team, 2024). We focus on GR00T because its backbone–head split is explicit and its checkpoints are public.

**Activation interpretability.** Linear probes recover task-relevant subspaces (Alain and Bengio, 2017); sparse autoencoders (Cunningham et al., 2023; Bricken et al., 2023; Templeton et al., 2024) factor activations into monosemantic features. NLA (Fraser-Taliente et al., 2026) complements these by producing *free-text* explanations  $y = \text{AV}(\mathbf{h})$  and a faithfulness anchor  $\hat{\mathbf{h}} = \text{AR}(y)$ . `nla-groot` ports the recipe to a VLA backbone.

**Activation steering.** Activation editing has demonstrated controllable behavior in LMs via linear directions (Turner et al., 2023; Zou et al., 2023; Liu et al., 2024; Panickssery et al., 2024). In robotics, behavioral steering is rarer; closed-loop sim is the appropriate validation modality when the model is a policy. We treat AR’s output as a candidate steering vector and *measure* whether it changes downstream behavior on LIBERO.

**Robotic simulation benchmarks.** LIBERO (Liu et al., 2023) is a suite of MuJoCo-based manipulation tasks with public assets; GR00T ships posttrained checkpoints per LIBERO sub-suite (e.g. `libero_goal`). We use these as the closed-loop ground truth.

### 3 Method

#### 3.1 Data: extraction and labeling

Per-token backbone activations are captured by a forward hook on the GR00T `backbone_features` tensor, before the post-VL LayerNorm and the cross-attention to the action head (canonicalized in `src/nla/layer_spec.py`). For each LeRobot episode and step we record  $\mathbf{h}_i \in \mathbb{R}^{2048}$  at three position types (text, image-patch, image-summary; see Fraser-Taliente et al. (2026)). A multimodal teacher (GPT-class) is shown the raw camera frame(s), instruction, and token-position metadata, and emits a structured bullet-point caption that we treat as gold for SFT.

A per-corpus scalar  $\alpha$  is estimated as the 75th percentile of  $\|\mathbf{h}\|_2$  and used for both volumetric calibration of injection and as the unit of MSE in AR training. For the LIBERO 4-suite combined corpus  $\alpha \approx 203.98$ .

#### 3.2 Architecture: AV and AR

Both AV and AR are LoRA fine-tunes (rank 32,  $\alpha = 64$ , full Qwen attention/MLP target set, bfloat16) of Qwen/Qwen3-4B-Instruct-2507.

**AV: activation  $\rightarrow$  text.** The prompt template contains a single `<|act_slot|>` token. At forward time we tokenize the prompt, embed the IDs, then overwrite the slot embedding with  $\alpha \cdot \text{normalize}(\mathbf{W}_p \mathbf{h})$  where  $\mathbf{W}_p \in \mathbb{R}^{h \times 2048}$  is a learnable projector. Loss is cross-entropy on the gold bullet caption.

**AR: text  $\rightarrow$  activation.** The base LM is physically truncated to its first 16 layers (matching GR00T’s `SELECT_LAYER`). A linear head reads the last non-pad hidden state of the prompt “*Summary of the following text: <text> <summary>*” and predicts  $\hat{\mathbf{h}}/\alpha \in \mathbb{R}^{2048}$ . Training loss is mean-squared error in  $\alpha$ -scaled space, plus an InfoNCE contrastive term with mined hard negatives (top- $k$  cosine,  $k=4$ ).

#### 3.3 Training

**Joint SFT.** AV and AR are updated on the same batches, so AR sees AV’s gold-text targets and AV cannot drift independently. The headline recipe (run `libero_4suite_v3`) uses 15,000 steps at batch = 4, lr =  $10^{-4}$ , 500-step warmup, cosine schedule, `ar_av_mix_max` = 0.4 (scheduled AR-on-AV-sampled-text mixing past 30% of training to close the train-deploy text gap), and held-out fraction = 0.05 episode-split.

**Optional GRPO phase.** After SFT, AV can be further trained by GRPO with the frozen AR as the reward model: for each  $\mathbf{h}$  we sample  $K$  captions, score each by  $-\|\mathbf{h}/\alpha - \text{AR}(y)\|^2$ , and apply a within-group advantage policy gradient with a Schulman-k3 KL to a frozen reference AV.

#### 3.4 Steering: language $\rightarrow$ live policy

At inference, a user supplies a short bullet caption  $y$ . We compute  $\hat{\mathbf{h}} = \text{AR}(y)$  and inject it into the live GR00T policy server by replacing or blending the backbone token features at one of three placements: a single `image_patch` token, *all* image-patch tokens (`image_patch_all`), or the image-summary token. A scalar `blend`  $\in [0, 2]$  controls intensity. The injection is wired through

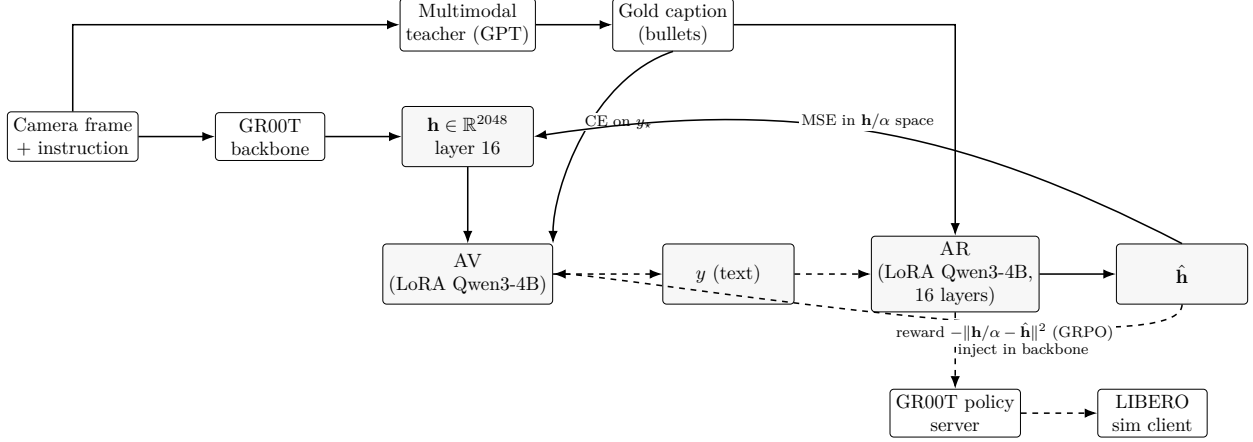


Figure 1: Data flow in `nla-groot`. Solid arrows are training paths; dashed arrows are inference / steering paths. AR doubles as the GRPO reward model and as the source of the live backbone steering vector.

`src/nla/steering/sim_policy_wrapper.py` so the official LIBERO sim client sees no difference apart from modified backbone features on every `get_action` call.

## 4 Evaluation harness

We deliberately decline to call any one number “the score”. Instead, each checkpoint is graded on three axes; a single scorecard (`scripts/eval/build_v3_scorecard.py`) emits PASS/WARN/FAIL per metric against pre-registered thresholds.

### 4.1 Axis 1 – Offline reconstruction and discriminability

- **Closed-loop cosine:**  $\cos(\mathbf{h}, \text{AR}(\text{AV}(\mathbf{h})))$  (rather than teacher-forced), to penalize AV captions that AR cannot invert.
- **Retrieval margin** and **retrieval@K**: matched vs. cross-pair cosine on  $\text{AR}(\text{AV}(\mathbf{h}))$  against a held-out pool; catches non-discriminative  $\hat{\mathbf{h}}$  (template collapse).

### 4.2 Axis 2 – Multimodal LLM judge

`scripts/eval/llm_judge_av_captions.py` re-renders the original frame, hands the multimodal judge the *frame* + caption, and scores three rubrics: (A) *format/length*, (B) *grounding* (does the caption describe what is in the picture, not a generic template), and (C) *appropriateness* (is it the right kind of caption for this position type). Gold (teacher) captions on the same frames give a calibration ceiling; we report both gold and AV pass rates and their gap.

### 4.3 Axis 3 – Closed-loop steerability A/B

Hardware-grounded: a steerability A/B (`scripts/eval/closed_loop_sim_ab.py` and `scripts/eval/steerability_eval.py`) runs LIBERO Goal episodes on the official GR00T checkpoint under three caption arms per task: *baseline* (no steer), *correct* (caption that matches the scene/instruction), and *wrong* (a contradictory caption). The headline number is

Table 1: V3 scorecard (`libero_4suite_v3`). Thresholds are pre-registered in `build_v3_scorecard.py`. Required metrics determine the *overall* verdict; the overall verdict is **FAIL**.

| Metric                                        | Value | Pass-thr | Warn-thr | Verdict |
|-----------------------------------------------|-------|----------|----------|---------|
| <b>Required</b>                               |       |          |          |         |
| <code>retrieval_margin</code>                 | 0.124 | 0.05     | 0.02     | PASS    |
| <code>judge_grounding_specific_pct</code>     | 0.417 | 0.55     | 0.40     | WARN    |
| <code>judge_anti_template_specific_pct</code> | 0.083 | 0.50     | 0.30     | FAIL    |
| <b>Informative</b>                            |       |          |          |         |
| <code>judge_appropriateness_pct</code>        | 1.000 | 0.80     | 0.60     | PASS    |
| <code>retrieval@1</code>                      | 0.039 | 0.25     | 0.15     | FAIL    |
| <code>retrieval@5</code>                      | 0.124 | 0.55     | 0.40     | FAIL    |
| <code>closed_greedy_cosine</code>             | 0.388 | 0.55     | 0.40     | FAIL    |
| <code>sim_correct_success</code>              | 0.000 | 0.30     | 0.15     | FAIL    |
| <code>sim_correct_minus_baseline_floor</code> | -1.00 | -0.10    | -0.30    | FAIL    |
| <code>sim_wrong_minus_baseline</code>         | -1.00 | 0.00     | 0.05     | PASS*   |
| <code>sim_correct_minus_wrong</code>          | 0.000 | 0.05     | 0.00     | WARN    |

\*Lower-is-better; trivially passes because *both* arms have zero success.

$\Delta_{cw} = \text{succ}_{\text{correct}} - \text{succ}_{\text{wrong}}$ ; we target  $\Delta_{cw} \geq +5\text{pp}$  as PASS. Per-object displacement and minimum end-effector distance are logged for failure analysis.

## 5 Experiments and results

### 5.1 Setup

The headline checkpoint, `libero_4suite_v3`, is trained on a combined LIBERO Goal/Spatial/Object/10 activation corpus with the hyperparameters of §3. Closed-loop sim is run against `nvidia/GROOT-N1.7-LIBERO/libero_goal` (embodiment `LIBERO_PANDA`) on the single task `put_the_bowl_on_the_plate`, three seeds  $\{0, 1, 2\}$ , `max_episode_steps=300`, eight action steps per chunk.

### 5.2 Scorecard

Table 1 reproduces the V3 scorecard verbatim from `data/sft/libero_4suite_v3/v3_scorecard.json`.

### 5.3 AV captions vs. teacher (judge)

Table 2 reports judge pass rates on a 24-row held-out sample (verbatim from `av_metrics.json`). The gap to gold is the honest signal: 25–50pp on grounding, 33–42pp on anti-template specificity. The gold ceiling itself is 75–83%, which sets an upper bound a workshop reviewer should expect a *credible* AV to approach.

### 5.4 Closed-loop steerability

Across three steer prompts  $\times$  two AR variants (`libero_goal_pilot` and `libero_4suite_v3`)  $\times$  3 seeds (Table 3, distilled from `data/eval/steerability_v1_vs_v3/metrics.json`), the baseline policy solves the task on every seed (`success_any_rate=1.0`) and every steering condition *stops* it from doing so (`success_any_rate=0.0`). This is a robust behavioral effect—steer-on does change behavior—but it

Table 2: Multimodal judge: gold (teacher) vs. AV captions, pass rates on a 24-row held-out sample per dataset. Numbers from `data/eval/steerability_v1_vs_v3/av_metrics.json`.

| AR                         | Held-out | Grounding | Approp. | Template-distinguish. |
|----------------------------|----------|-----------|---------|-----------------------|
| libero_goal_pilot          | gold     | 0.833     | 0.917   | 0.833                 |
|                            | AV       | 0.333     | 0.667   | 0.500                 |
| libero_4suite_v3           | gold     | 0.833     | 1.000   | 0.833                 |
|                            | AV       | 0.583     | 1.000   | 0.500                 |
| libero_4suite_v3 @ 4-suite | gold     | 0.750     | 1.000   | 0.500                 |
|                            | AV       | 0.417     | 1.000   | 0.083                 |

Table 3: LIBERO Goal closed-loop steerability on `put_the_bowl_on_the_plate`, three seeds. “Target winner rate” is the fraction of seeds in which the maximum-displaced tracked body was the one named by the steer prompt.

| Condition                | succ <sub>any</sub> | Target displacement (m) | Target winner rate |
|--------------------------|---------------------|-------------------------|--------------------|
| baseline                 | 1.00                | 0.145 (bowl)            | 1.00               |
| steer_bowl_plate (v1 AR) | 0.00                | 0.072 (bowl)            | 1.00               |
| steer_wine Rack (v1 AR)  | 0.00                | 0.066 (bottle)          | 0.00               |
| steer_bowl_stove (v1 AR) | 0.00                | 0.077 (bowl)            | 1.00               |
| steer_bowl_plate (v3 AR) | 0.00                | 0.069 (bowl)            | 0.67               |
| steer_wine Rack (v3 AR)  | 0.00                | 0.070 (bottle)          | 0.00               |
| steer_bowl_stove (v3 AR) | 0.00                | 0.072 (bowl)            | 1.00               |

is symmetric: the correct (`bowl_plate`) and contradictory (`wine Rack`, `bowl_stove`) prompts both block the task, giving  $\Delta_{cw} = 0$  and disqualifying any semantic claim from this pilot. The *per-object* displacement and end-effector traces (Figure 2) reveal that all steered conditions collapse the bowl’s per-object displacement to  $\approx 0.07\text{m}$  versus  $0.145\text{m}$  at baseline—i.e. steering reduces motion uniformly rather than redirecting it to the prompted object.

## 5.5 Reading the results

Three observations dominate.

**(i) Reconstruction passes; behavior does not.** The retrieval margin and appropriateness rate would, on their own, read as a successful training run. The judge’s grounding and anti-template scores, and the symmetric LIBERO-Goal failure, contradict that read directly. This validates the harness: had we shipped only the reconstruction-style metrics we would have over-claimed.

**(ii) Shorthand template collapse persists in V3.** The 8.3% anti-template rate on the 4-suite holdout means AV captions are mostly indistinguishable from a corpus-mean template. This is the same failure mode reported for the earlier `droid_100ep_v2_nce` run (V2 post-mortem, `docs/sft_plan/06_v2_postmortem_v3_rerun.md`); V3’s mitigations (`ar_av_mix_max`, InfoNCE hard negatives, episode-split) narrowed the gap but did not close it.

**(iii) Steering bites, but is not yet semantic.** Every steered condition blocks task completion—so there is energy in the perturbation—but the prompt’s identity does not appear to choose *what*



Figure 2: Per-tracked-body summaries across conditions on `put_the_bowl_on_the_plate`. Baseline (left bar in each group) is the only condition with task-typical bowl displacement; every steering arm dampens motion approximately uniformly.

the policy attends to. We attribute this to (a) the AR’s train–deploy distribution gap (it is trained on gold-style bullet captions and deployed on hand-written ones), and (b) the `image_patch_all` placement at `blend = 1.0` being a coarse intervention (consistent with the open-loop probe in `scripts/eval/nla_steer_quant_probe.py`).

## 6 Limitations

- **Single task, small  $n$ .** The closed-loop sim numbers above are three seeds on one LIBERO Goal task. We frame them as feasibility, not a benchmark; a multi-task LIBERO sweep with per-prompt affordance scoring is required before any quantitative semantic-steering claim.
- **Teacher– $\mathbf{h}$  confound.** The multimodal teacher sees the original frame and may mention scene facts that are only weakly present in  $\mathbf{h}$ , so SFT loss does not strictly equal “what  $\mathbf{h}$  encodes”. This is inherited from the NLA recipe and motivates the judge’s grounding axis.
- **AR train–deploy gap.** AR is trained on gold-style bullet captions; user-written steer prompts are out of distribution. The `ar_av_mix_max` schedule reduces but does not eliminate this gap.
- **Single backbone, single layer.** Activations are extracted at layer 16 (`SELECT_LAYER`) before the post-VL LayerNorm. Other choices (`vl_embeds`, DiT mid-block) are documented in `src/nla/layer_spec.py` but not ablated here.
- **Judge as oracle.** The grounding judge is itself a large multimodal model and inherits its own miscalibrations; we mitigate by pairing every AV score with a gold score on identical frames.

## 7 Discussion and outlook

`nla-groot` establishes that the NLA recipe is implementable on a production VLA backbone and that the resulting checkpoint can be put in the loop of a public closed-loop simulator. The principal contribution of this short paper is the harness, not the headline number: a multi-axis scorecard with pre-registered thresholds and a behavioral arm is, in our experience, what separates an apparently successful run from an actually successful one.

A natural next step is the GRPO phase against the AR-as-reward signal, which we have plumbed (`scripts/training/run_grpo.py`) but do not report here. We expect GRPO with judge-blended reward to attack the template-collapse mode head-on, since judge rewards penalize generic captions in a way the supervised cross-entropy does not.

## Reproducibility

Code: <https://github.com/<anonymized>/nla-groot> (commit pinned at submission). Pinned environment, full hyperparameters and canonical commands appear in App. A. Seeds: SFT and

steerability rollouts both default to `seed=0`; the closed-loop sim numbers reported here use seeds  $\{0, 1, 2\}$ .

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## A Reproducibility appendix

**Hardware.** A single NVIDIA H100 (80GB) is sufficient for joint SFT at `batch_size=4`, `grad_accum_steps=1`, `gradient_checkpointing=True`. LIBERO sim is CPU-rendered (MUJOCO\_GL=osmesa).

**Software.** The pinned package set used for the reported runs is in `paper/repro/requirements.txt`. Notable versions: `torch==2.7.1+cu128`, `transformers==4.57.3`, `peft==0.17.1`, `accelerate==1.13.0`, `datasets==3.6.0`, `deepspeed==0.17.6`, `safetensors==0.7.0`, `numpy==1.26.4`, Python 3.10.12.

**Checkpoint identifiers.** Backbone: `nvidia/GR00T-N1.7-3B` (canonical; we extract from the LIBERO-finetuned variants `nvidia/GR00T-N1.7-LIBERO/libero_{suite}`). Base LM for AV/AR: `Qwen/Qwen3-4B-Instruct-2507`.

**Canonical commands.** See `paper/repro/canonical_commands.sh` for the SFT, GRPO, extraction, scorecard, and steerability invocations used in this paper.

**Hyperparameters.** The full SFT config snapshot is at `data/sft/libero_4suite_v3/config.json`; we reproduce its non-default entries in Table 4.

Table 4: Non-default SFT hyperparameters for `libero_4suite_v3`.

| Key                                           | Value      |
|-----------------------------------------------|------------|
| <code>alpha</code>                            | 203.977    |
| <code>batch_size</code>                       | 4          |
| <code>grad_accum_steps</code>                 | 1          |
| <code>learning_rate</code>                    | $10^{-4}$  |
| <code>warmup_steps</code>                     | 500        |
| <code>total_steps</code>                      | 15,000     |
| <code>av_weight / ar_weight</code>            | 1.0 / 1.0  |
| <code>ar_contrastive_weight</code>            | 0.5        |
| <code>ar_nce_hard_negatives_per_anchor</code> | 4          |
| <code>nce_temperature</code>                  | 0.1        |
| <code>ar_av_mix_max</code>                    | 0.4        |
| <code>ar_av_mix_warmup_frac</code>            | 0.3        |
| <code>gradient_checkpointing</code>           | true       |
| <code>eval_closed_loop</code>                 | true       |
| <code>closed_loop_temperatures</code>         | {0.0, 0.7} |
| <code>balance_position_mix</code>             | true       |
| <code>split_by</code>                         | episode    |
| <code>held_out_fraction</code>                | 0.05       |

**Scorecard schema.** Per-metric thresholds and JSON sources are defined in `scripts/eval/build_v3_scorecard.py`; the machine-readable scorecard for this paper is `data/sft/libero_4suite_v3/v3_scorecard.json`.